

0. Course Details & Introduction

COMP3314
Introduction to Machine Learning

Dong Xu



But now Tesla and xAI founder Elon Musk has come out and said traditional coding may be dead by the end of this year.

"I think actually things will move maybe even by the end of this year to where you don't even bother doing coding. The AI just creates the binary directly and the AI can create a much more efficient binary than can be done by any compiler (a type of computer program that converts code from one programming language into another).

"So just say create the optimised binary for this particular outcome and you actually bypass even traditional coding. That's an intermediate step that actually will not be needed probably by... I'd say the end of this year," Musk opined at a conference.

Entry-level coding jobs have indeed dried up as companies increasingly rely on these tools. A report published last month by Revelio Labs and the World Economic Forum showed a 35% drop in entry-level job postings; software development and programming jobs were among the hardest hit.

Last year, Anthropic co-founder and CEO Dario Amodei had said AI could eliminate half of all entry-level white-collar jobs and cause unemployment to rise to 10% to 20%.

https://www.ndtv.com/artificial-intelligence/elon-musk-predicts-coding-jobs-will-be-obsolete-by-end-of-this-year-due-to-ai-advancements-11612260

Course Staff

- Instructors
  - Prof. Dong Xu
    - Email: dongxu@cs.hku.hk
    - Consultation hours: Tuesday 4:30pm - 5:30pm & by appointment
- TA
  - Qiyuan Qiao
    - Email: qiaoqy@connect.hku.hk
    - Consultation hours: Thursday 4:30pm - 5:30pm & by appointment
  - Junjie Zhang
    - Email: u3014222@connect.hku.hk
    - Consultation hours: Friday 4:30pm - 5:30pm & by appointment
  - Wei Dai
    - Email: u3014487@connect.hku.hk
    - Consultation hours: Wednesday 4:30pm - 5:30pm & by appointment

Schedule

#	Date	Venue	Time	Topic*
1	7 Sep	MB167	3pm - 550pm	Course Details / KNN Clustering, Regression
2	14 Sep	MB167	3pm - 550pm	Support Vector Machine & Performance Evaluation
3	21 Sep	MB167	3pm - 550pm	Boosting & Decision Tree
4	28 Sep	MB167	3pm - 550pm	Clustering & Applications
5	5 Oct	MB167	3pm - 550pm	Dimension Reduction
6	12 Oct	MB167	3pm - 550pm	Reading week (no class)
7	19 Oct	MB167	3pm - 550pm	Public holiday (no class)
8	26 Oct	MB167	3pm - 550pm	Perceptron & Adaline
9	2 Nov	MB167	3pm - 550pm	Logistic Regression
10	9 Nov	MB167	3pm - 550pm	Neural Networks and Backpropagation
11	16 Nov	MB167	3pm - 550pm	Student Presentation
12	23 Nov	MB167	3pm - 550pm	Student Presentation
13	30 Nov	MB167	3pm - 550pm	Course Review, Introduction to Machine learning Software and Assignment Answers

Assessment

- One assignment (10%)
- Indivisual project (10%)
- Group project (30%)
  - Read a machine learning research paper
  - Presentation to class + Q&A
  - About 30 Groups
  - 3~4 students per group depending on the total number of students choosing this class
- Final examination (50%): written, closed book, 2 hours

List of Machine Learning Papers (1)

- [1] S. Yan, **D. Xu**, B. Zhang, H. Zhang, Q. Yang and S. Lin, "Graph Embedding and Extensions: A General Framework for Dimensionality Reduction," *IEEE Transactions on Pattern Analysis and Machine Intelligence* (T-PAMI), 29(1), pp. 40-51, January 2007.
- [2] S. Yan, **D. Xu**, Q. Yang, L. Zhang, X. Tang and H. Zhang, "Multilinear Discriminant Analysis for Face Recognition," *IEEE Transactions on Image Processing* (T-IP), 16(1), pp. 212-220, January 2007.
- [3] **D. Xu**, S. Yan, D. Tao, S. Lin and H. Zhang, "Marginal Fisher Analysis and Its Variants for Human Gait Recognition and Content-Based Image Retrieval," *IEEE Transactions on Image Processing* (T-IP), 16(11), pp. 2811-2821, November 2007.
- [4] **D. Xu**, S. Yan, L. Zhang, S. Lin, H. Zhang and T. S. Huang, "Reconstruction and Recognition of Tensor-based Objects with Concurrent Subspaces Analysis," *IEEE Transactions on Circuits and Systems for Video Technology* (T-CSVT), 18(1), pp. 36-47, January 2008.
- [5] F. Nie*, **D. Xu**, I. W. Tsang and C. Zhang, "Flexible Manifold Embedding: A Framework for Semi-supervised and Unsupervised Dimension Reduction," *IEEE Transactions on Image Processing* (T-IP), 19(7), pp. 1921-1932, July 2010.
- [6] Y. Yang*, **D. Xu**, F. Nie*, S. Yan and Y. Zhuang, "Image Clustering using Local Discriminant Models and Global Integration," *IEEE Transactions on Image Processing* (T-IP), 19(10), pp. 2761-2773, October 2010.

List of Machine Learning Papers (2)

- [7] F. Nie*, Z. Zeng*, I. W. Tsang**, D. Xu** and C. Zhang, "Spectral Embedded Clustering: A Framework for In-Sample and Out-of-Sample Spectral Clustering," *IEEE Transactions on Neural Networks* (T-NN), 22(11), pp. 1796-1808, November 2011.
- [8] L. Duan*, I. W. Tsang and **D. Xu**, "Domain Transfer Multiple Kernel Learning," *IEEE Transactions on Pattern Analysis and Machine Intelligence* (T-PAMI), 34(3), pp. 465-479, March 2012.
- [9] L. Duan*, **D. Xu** and I. W. Tsang, "Domain Adaptation from Multiple Sources: A Domain-Dependent Regularization Approach," *IEEE Transactions on Neural Networks and Learning Systems* (T-NNLS), 23(3), pp. 504-518, March 2012.
- [10] L. Chen*, I. W. Tsang and **D. Xu**, "Laplacian Embedded Regression for Scalable Manifold Regularization," *IEEE Transactions on Neural Networks and Learning Systems* (T-NNLS), 23(6), pp. 902-915, June 2012.
- [11] L. Duan*, **D. Xu**, I. W. Tsang and J. Luo, "Visual Event Recognition in Videos by Learning from Web Data," *IEEE Transactions on Pattern Analysis and Machine Intelligence* (T-PAMI), 34(9), pp. 1667-1680, September 2012.
- [12] X. Xu*, W. Li* and **D. Xu**, "Distance Metric Learning using Privileged Information for Face Verification and Person Re-identification," *IEEE Transactions on Neural Networks and Learning Systems* (T-NNLS), 26(12), pp. 3150-3162, December 2015.

Prerequisite

- Programming
- Basic linear algebra, calculus and statistics
  - Students without strong programming capability and mathematics may spend a lot of time on this course

Assessment - Project - Group Selection

- Select your group on Moodle
  - Start: 7 Sep
  - Deadline: 17 Sep 23:55
  - We will randomly assign a group for you if you don’t make a selection by the deadline
- Member swapping
  - Allowed if approved before start of project
    - All group members of both groups must agree

Assessment - Project - Paper Selection

- Read a machine learning paper
- Implement / test the paper
  - May use CS GPU Farm
  - Employ own test/training data if possible
  - Make your code available to others via github
- List of project topics will be made available on 23 Sep
- Topic selection
  - Start on 18 Sep
  - Deadline: 24 Sep 23:59

Plagiarism

- What is Plagiarism ?
  - https://tl.hku.hk/plagiarism/
- If a student commits plagiarism, with evidence after investigation, no matter whether the student concerned admits or not, a penalty will be imposed
- First Attempt: the student shall be warned in writing and receive zero mark for the whole assignment or the whole test; if the student does not agree, s/he can appeal to the Programme Director within a week
- Subsequent Attempt: May impose any of the following penalties: a published reprimand, suspension of study for a period of time, fine, or expulsion from the University
- Both the student who copies and the student who offers his/her work for copying shall be penalized

Course Materials

- Available on Moodle
  - Use the provided materials responsibly
  - For your own research and private study only
  - Don’t distributed to others without the appropriate authorization
    - We do not hold copyright for most materials

References and Acknowledgements

- Textbooks
  - Python Machine Learning (PML), Sebastian Raschka & Vahid Mirjalili
  - Introduction to Machine Learning (IML), Ethem Alpaydin
  - Pattern Recognition and Machine Learning, Christopher Bishop
  - Pattern Classification, Richard Duda, Peter Hart, and David Stork

- Other Courses
  - Machine Learning Crash Course, Google
  - Machine Learning, Coursera
  - CS229, Berkeley
- Additional references (if any) will be listed on the lecture slides

Tools

- Provided examples and exercises will use the following tools
  - Python 3
  - NumPy
  - SciPy library
  - scikit-learn
  - Matplotlib
  - Pandas
  - Tensorflow
  - Jupyter Notebook
- All of the above can be installed conveniently with Anaconda
  - We recommend to use Google CoLab

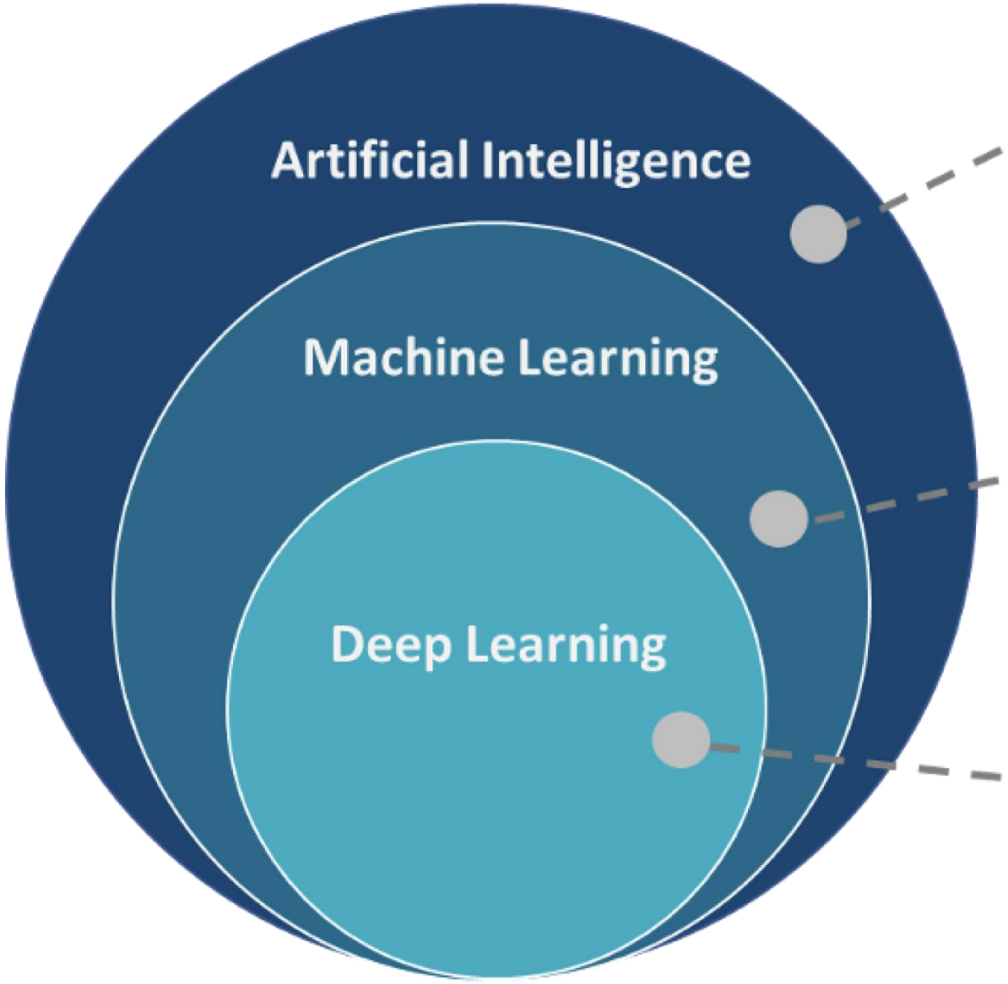
Discussion Forum

- If you have questions about course materials, please **use the forum**
  - Answer questions from other students!

Q & A

Introduction

What is AI? ML? DL?



Artificial Intelligence
A technique which enables machines to mimic human behaviour

Machine Learning
Subset of AI technique which uses statistical methods to enable machines to improve with experience.

Deep Learning
Subset of Machine Learning which make the computation of multi-layer neural network feasible

Machine Learning

- Principles, methods and algorithms for learning and prediction on the basis of past experience
  - Learn existing and known structures and rules or discover new findings and structures.
  - Already everywhere: speech recognition, hand-written character recognition, face recognition, information retrieval, operating systems, fraud detection, security, defense application.

Preprocessing: Feature extraction for document classification

Task: Classify the documents into predefined categories (*i.e.*, classes)

Label	Titles
B1	A Course on Integral Equations
B2	Attractors for Semigroups and Evolution Equations
B3	Automatic Differentiation of Algorithms: Theory, Implementation, and Application
B4	Geometrical Aspects of Partial Differential Equations
B5	Ideals, Varieties, and Algorithms – An Introduction to Computational Algebraic Geometry and Commutative Algebra
B6	Introduction to Hamiltonian Dynamical Systems and the N-Body Problem
B7	Knapsack Problems: Algorithms and Computer Implementations
B8	Methods of Solving Singular Systems of Ordinary Differential Equations
B9	Nonlinear Systems
B10	Ordinary Differential Equations
B11	Oscillation Theory for Neutral Differential Equations with Delay
B12	Oscillation Theory of Delay Differential Equations
B13	Pseudodifferential Operators and Nonlinear Partial Differential Equations
B14	Sinc Methods for Quadrature and Differential Equations
B15	Stability of Stochastic Differential Equations with Respect to Semi-Martingales
B16	The Boundary Integral Approach to Static and Dynamic Contact Problems
B17	The Double Mellin-Barnes Type Integrals and Their Applications to Convolution Theory

After the preprocessing step, each document is represented as one data point in the 16-D feature space.

TABLE 3
The 16 × 17 term-document matrix corresponding to the book titles in Table 2.

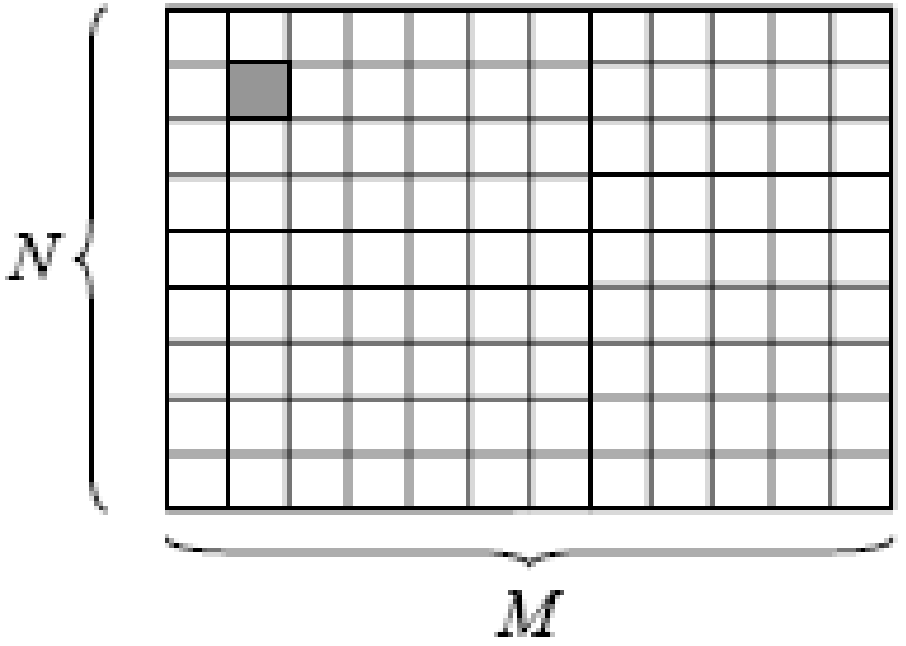
Terms	Documents																
	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12	B13	B14	B15	B16	B17
algorithms	0	0	1	0	1	0	1	0	0	0	0	0	0	0	0	0	0
application	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1
delay	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0
differential	0	0	0	1	0	0	0	1	0	1	1	1	1	1	1	0	0
equations	1	1	0	1	0	0	0	1	0	1	1	1	1	1	1	0	0
implementation	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0
integral	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1
introduction	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0
methods	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0
nonlinear	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0
ordinary	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0
oscillation	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0
partial	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0
problem	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	1	0
systems	0	0	0	0	0	1	0	1	1	0	0	0	0	0	0	0	0
theory	0	0	1	0	0	0	0	0	0	0	1	1	0	0	0	0	1

Intensity Image Representation

Grey-scale / intensity image:

- ***N* rows × *M* columns**

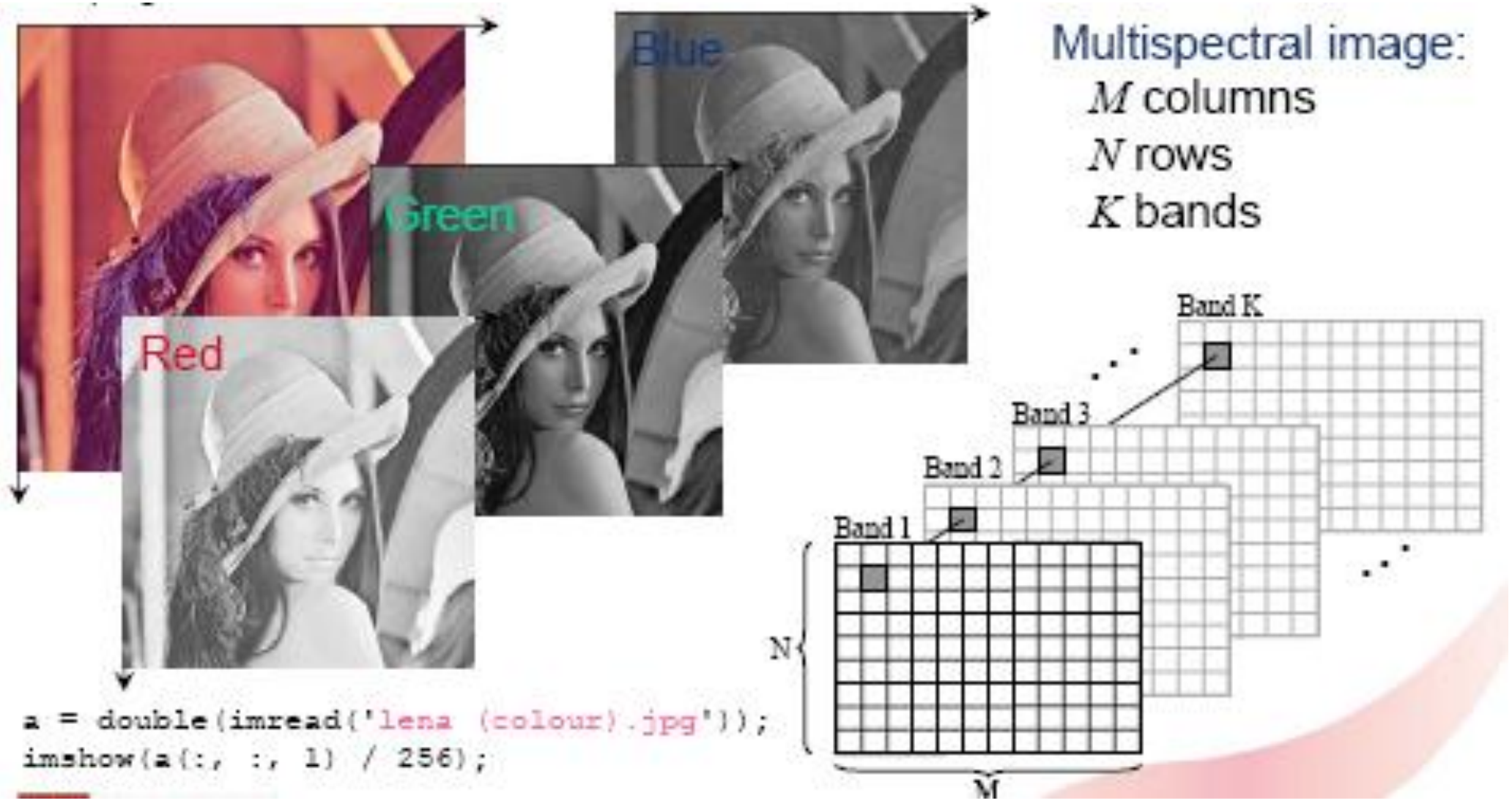




$$g(n,m)=\begin{bmatrix} g(1,1) & g(1,2) & \cdots & g(1,M) \\ g(2,1) & g(2,2) & \cdots & g(2,M) \\ \vdots & \vdots & \ddots & \vdots \\ g(N,1) & g(N,2) & \cdots & g(N,M) \end{bmatrix}$$

One intensity image is represented as *NM* data points (*i.e.*, pixels) in the 1-D feature space.

Color Image and Multi-spectral Image

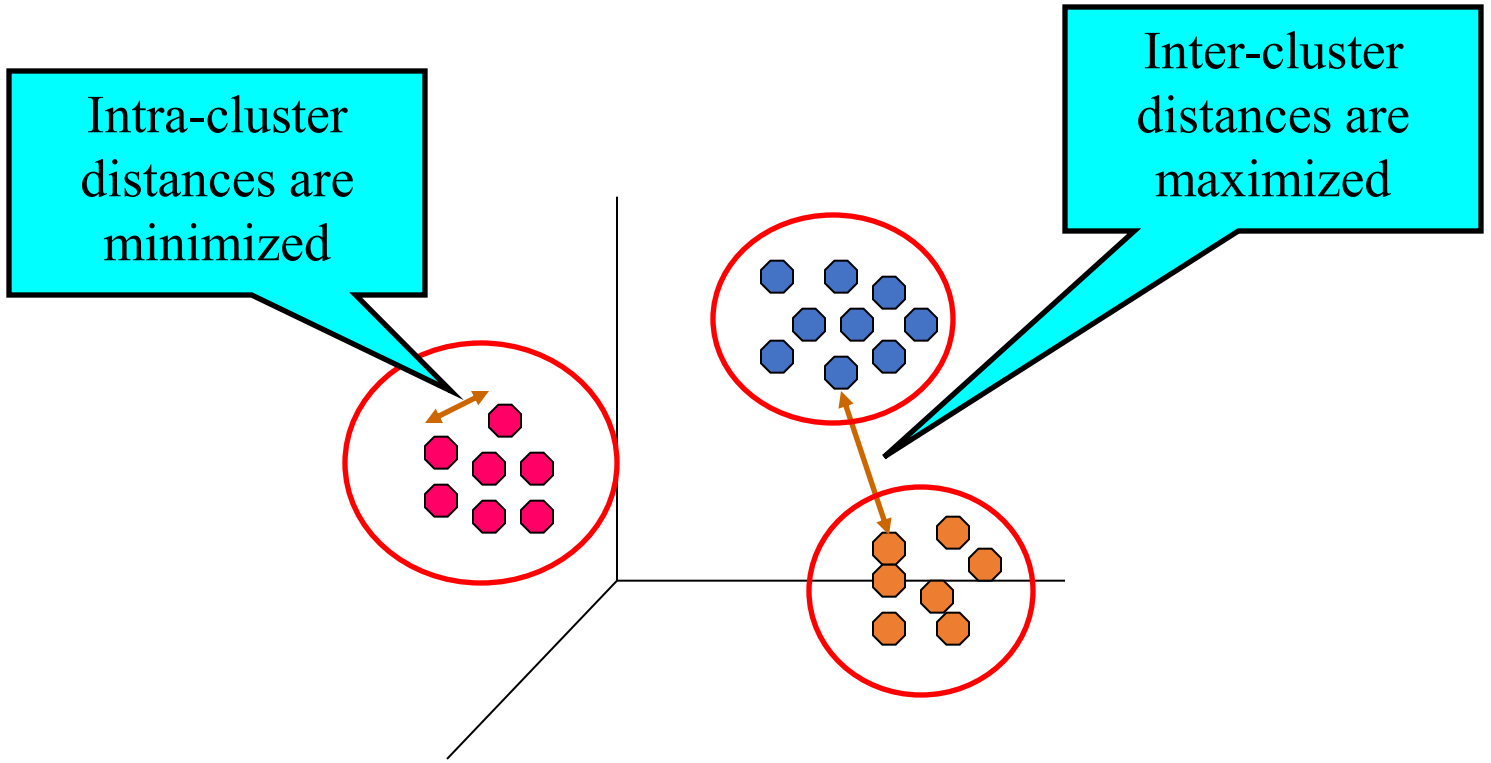


One color (*resp.* multi-spectral) image is represented as *NM* data points (*i.e.*, pixels) in the 3-D (*resp. K*-D) feature space.

Types of Learning Problems

- Supervised learning: where we get a set of training inputs and outputs
  - Regression
  - Classification
- Unsupervised learning: where we are interested in capturing inherent organization in the data
  - Clustering
  - Density estimation
- Semi-supervised learning: where we get part of training inputs and outputs, and part of data without label information
- Transfer learning: a large number of labeled training data in the source domain and a limited number of or even no labeled training data in the target domain
  - data distribution mismatch between the training data in the source domain and the testing data in the target domain

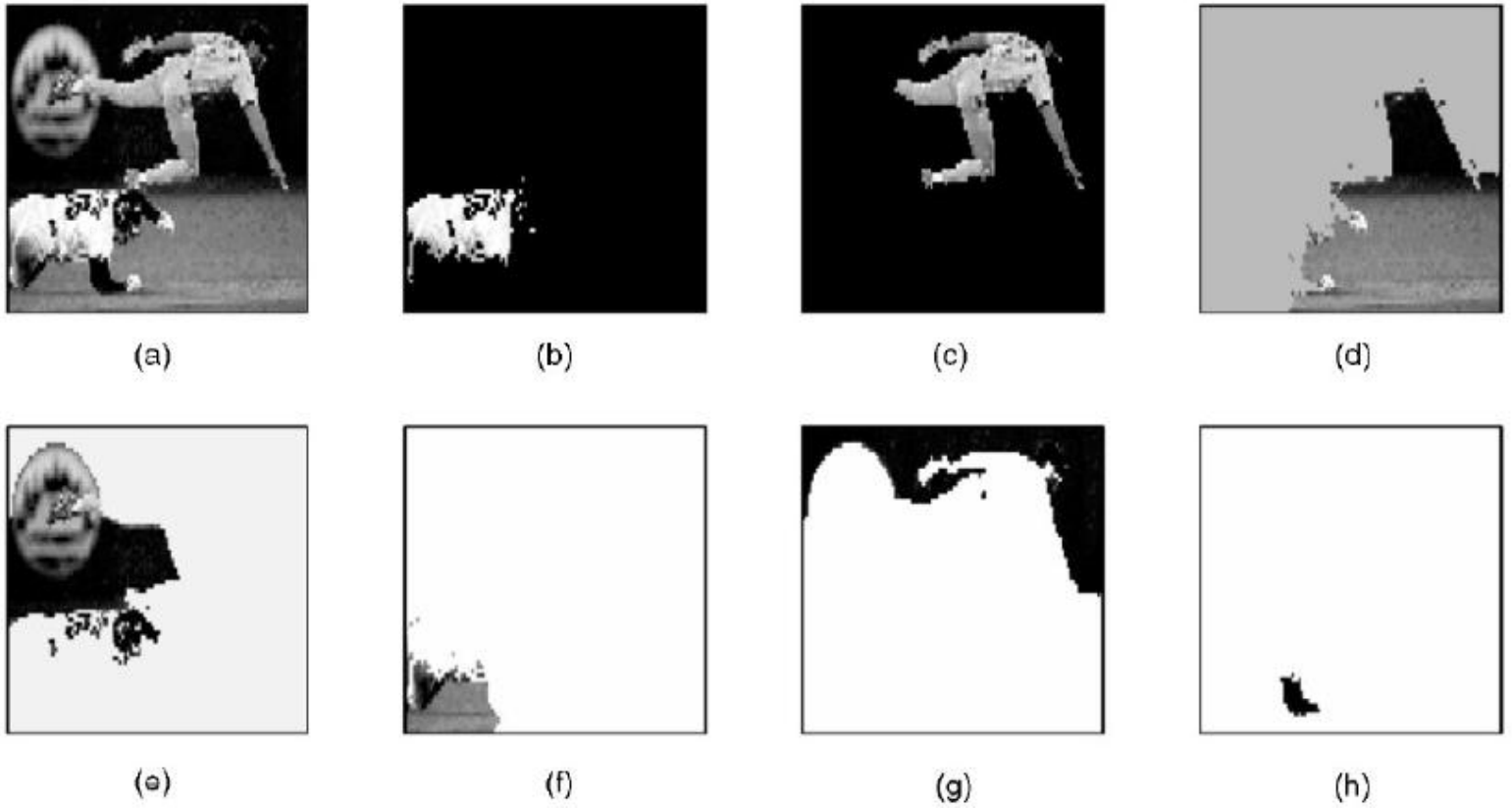
Clustering: Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups



Clustering in Image Segmentation

- Data reduction - obtain a compact representation for *interesting* image data in terms of a set of components
- Find components that belong together (form clusters)



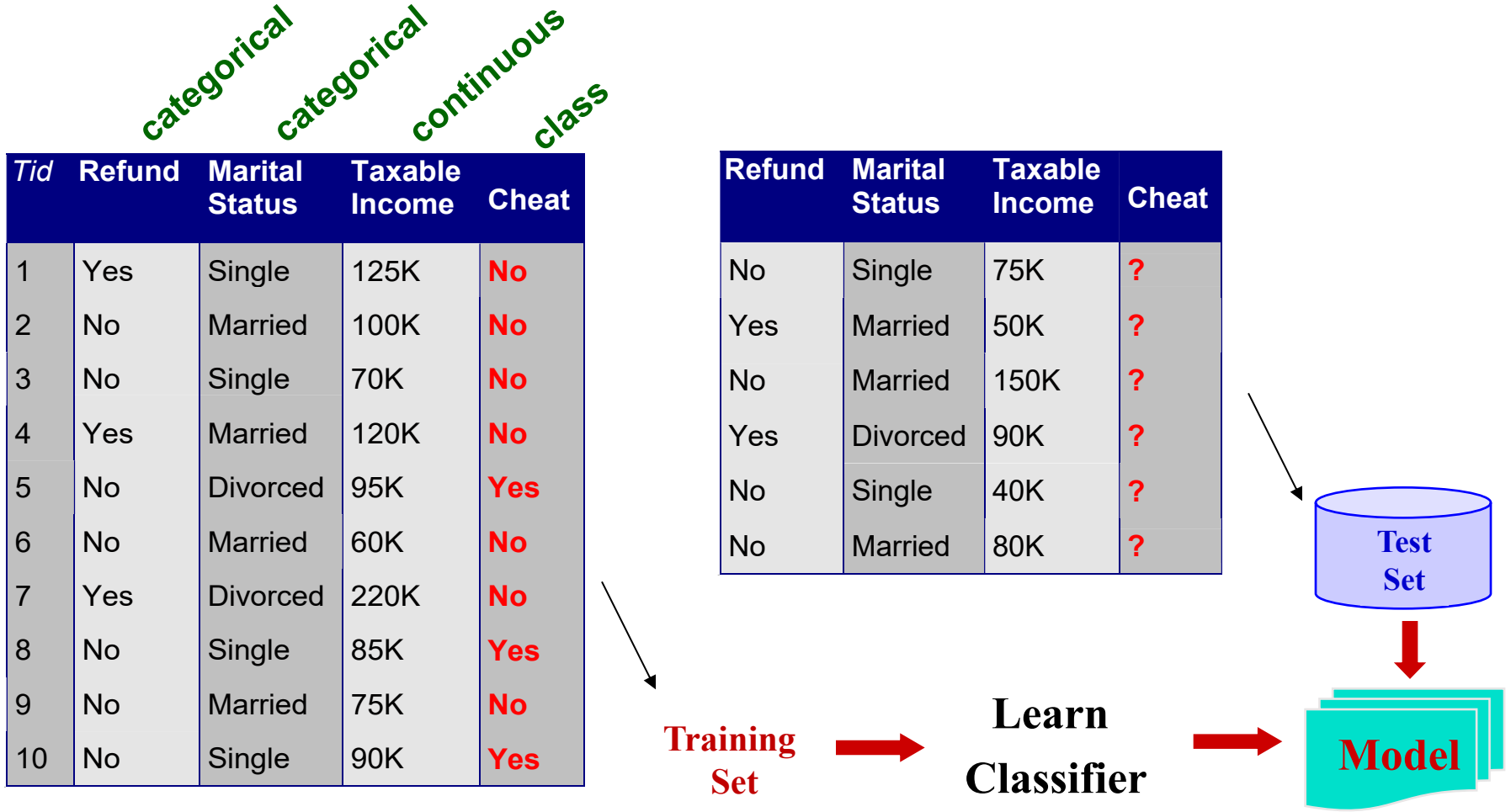


From: *Normalize Cut Normalized Cuts and Image Segmentation*, Jianbo Shi and Jitendra Malik, T-PAMI 2000

Classification

- Given a collection of records (*training set*)
  - Each record contains a set of *attributes*, one of the attributes is the *class*.
- Find a *model* for class attribute as a function of the values of other attributes.
- Goal: previously unseen records should be assigned a class as accurately as possible.
  - A *test set* is used to determine the accuracy of the model. Usually, the given data set is divided into training and test sets, with training set used to build the model and test set used to validate it.

Classification Example



Face Recognition in Surveillance



What makes face recognition hard?

Expression



What makes face recognition hard?

Lighting



What makes face recognition hard?

Occlusion

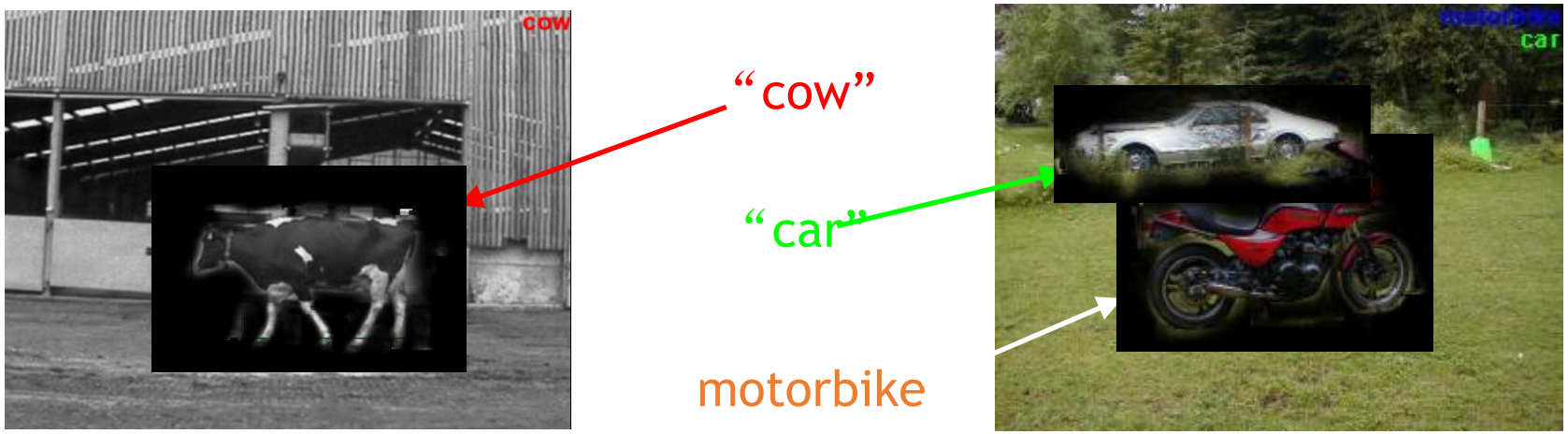


What makes face recognition hard?

Viewpoint



Object Recognition



- Different levels of recognition
  - *Which* object class is in the image? ⇒ Obj/Img categorization
  - *Where* is it in the image? ⇒ Detection/Localization
  - *Where exactly* — which pixels? ⇒ F/G segmentation

Challenges 1: view point variation



Michelangelo Italy 1475-1564

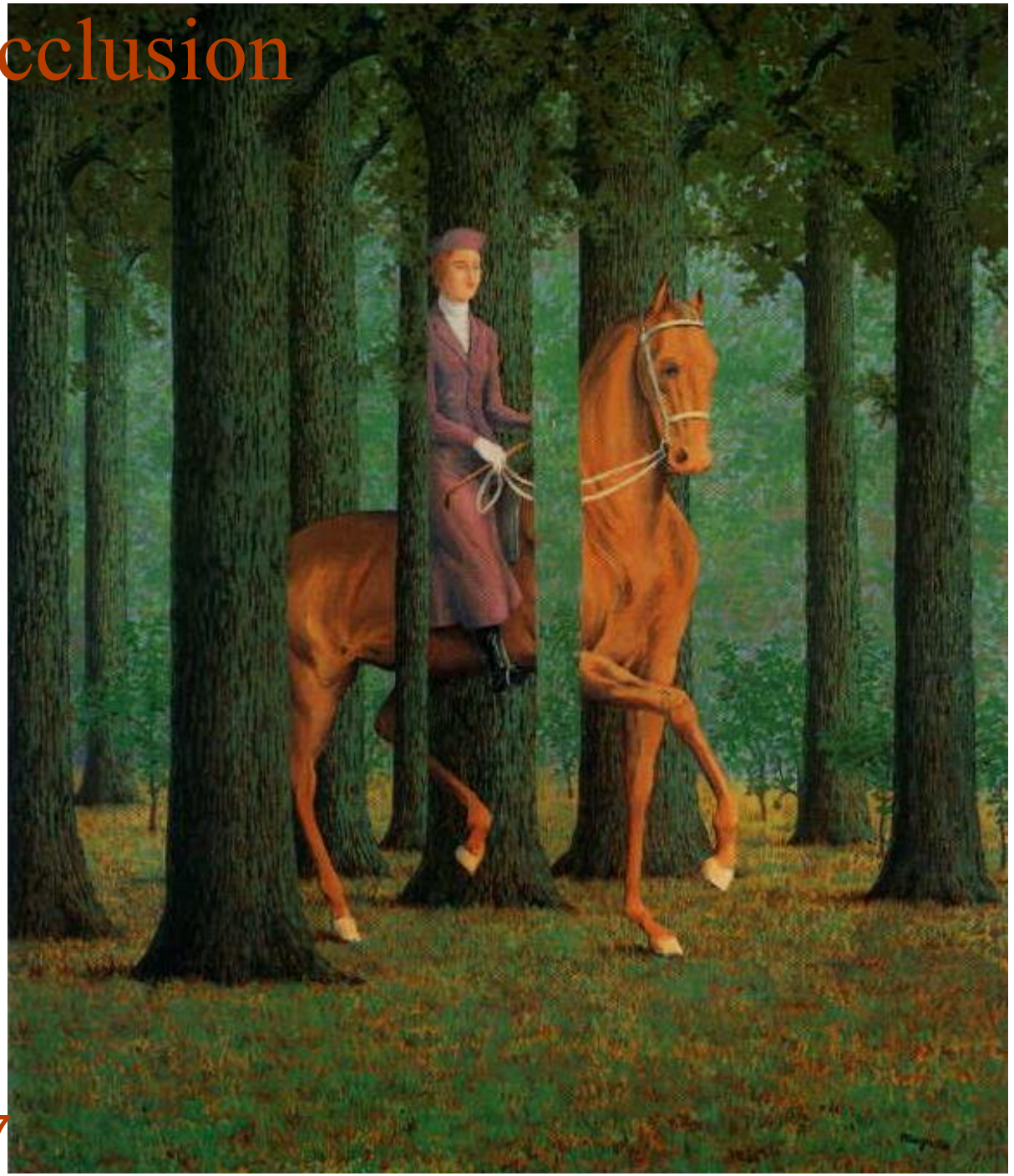
Challenges 2: illumination





slide credit: S. Ullman

Challenges 3: occlusion

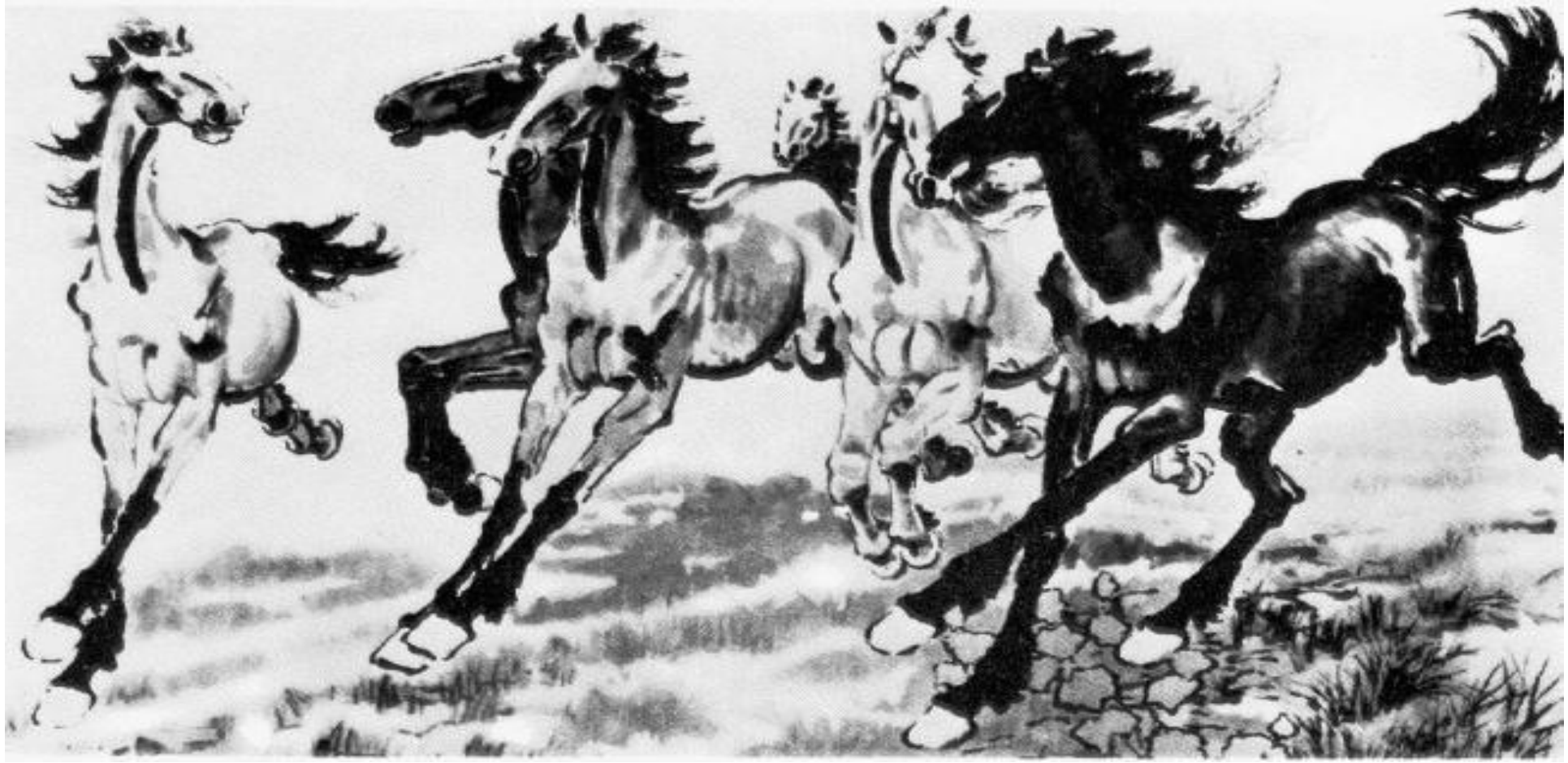


Magritte, 1957

Challenges 4: scale



Challenges 5: deformation



Xu, Beihong 1943

Challenges 6: background clutter



Klimt, 1913

Challenges 7: intra-class variation





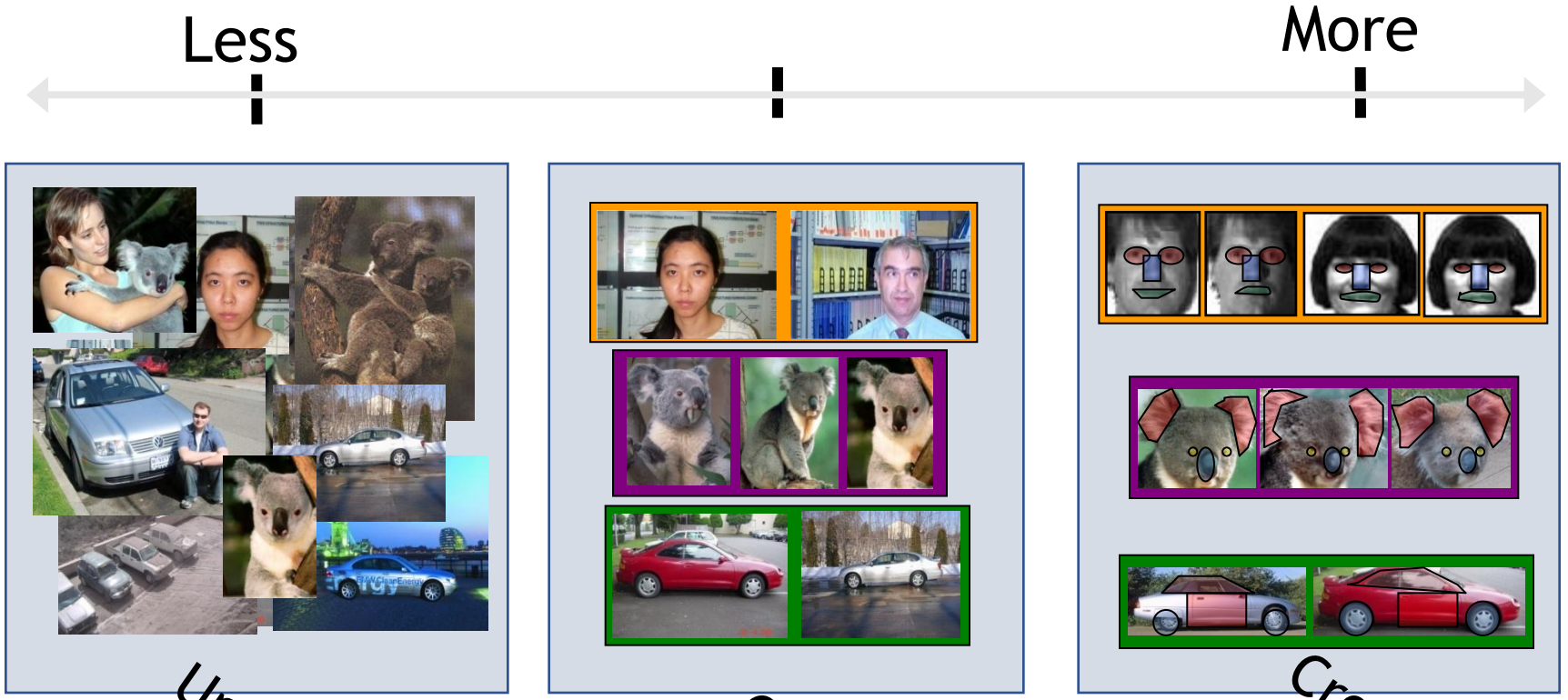








learning with minimal supervision



History: early object

History: early object categorization








~10,000 to 30,000

What could be done with recognition algorithms?

There is a wide range of applications, including…



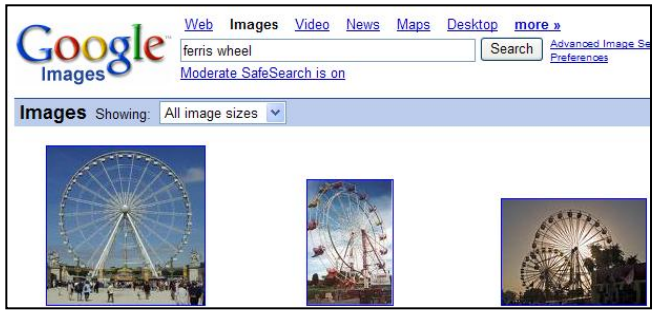
Autonomous robots

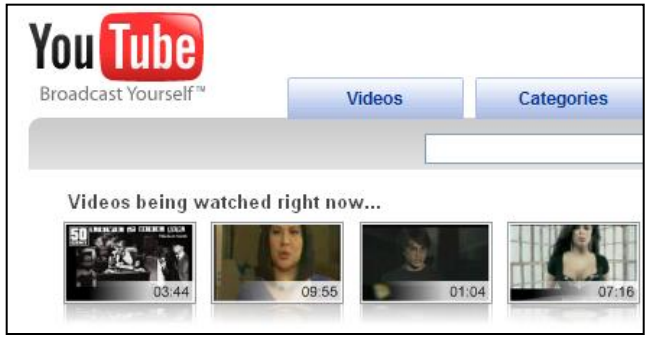


Navigation, driver safety



Landmark search





Content-based retrieval and analysis for images and videos



Medical image analysis

History of AI



History of AI - Turing Award Winners

- Marvin Minsky (1969)
- John McCarthy (1971)
- Edward Feigenbaum and Raj Reddy (1994)
- Judea Pearl (2011)
- Yoshua Bengio, Geoffrey Hinton, and Yann LeCun (2019)

https://awards.acm.org/about/2018-turing#:~:text=Geoffrey%20Hinton,-Backpropagation%3A%20In%20a&text=Boltzmann%20Machines%3A%20In%201983%2C%20with,of%20the%20input%20or%20output

Geoffrey Hinton

- Backpropagation: In a 1986 paper, "Learning Internal Representations by Error Propagation," co-authored with David Rumelhart and Ronald Williams, Hinton demonstrated that the backpropagation algorithm allowed neural nets to discover their own internal representations of data, making it possible to use neural nets to solve problems that had previously been thought to be beyond their reach. The backpropagation algorithm is standard in most neural networks today.
- Boltzmann Machines: In 1983, with Terrence Sejnowski, Hinton invented Boltzmann Machines, one of the first neural networks capable of learning internal representations in neurons that were not part of the input or output.
- Improvements to convolutional neural networks: In 2012, with his students, Alex Krizhevsky and Ilya Sutskever, Hinton improved convolutional neural networks using rectified linear neurons and dropout regularization. In the prominent ImageNet competition, Hinton and his students almost halved the error rate for object recognition and reshaped the computer vision field.

Yann LeCun

- Convolutional neural networks: In the 1980s, LeCun developed convolutional neural networks, a foundational principle in the field, which, among other advantages, have been essential in making deep learning more efficient. In the late 1980s, while working at the University of Toronto and Bell Labs, LeCun was the first to train a convolutional neural network system on images of handwritten digits. Today, convolutional neural networks are an industry standard in computer vision, as well as in speech recognition, speech synthesis, image synthesis, and natural language processing. They are used in a wide variety of applications, including autonomous driving, medical image analysis, voice-activated assistants, and information filtering.
- Improving backpropagation algorithms: LeCun proposed an early version of the backpropagation algorithm (backprop), and gave a clean derivation of it based on variational principles. His work to speed up backpropagation algorithms included describing two simple methods to accelerate learning time.
- Broadening the vision of neural networks: LeCun is also credited with developing a broader vision for neural networks as a computational model for a wide range of tasks, introducing in early work a number of concepts now fundamental in AI. For example, in the context of recognizing images, he studied how hierarchical feature representation can be learned in neural networks—a concept that is now routinely used in many recognition tasks. Together with Léon Bottou, he proposed the idea, used in every modern deep learning software, that learning systems can be built as complex networks of modules where backpropagation is performed through automatic differentiation. They also proposed deep learning architectures that can manipulate structured data, such as graphs.

Yann LeCun

- Yann LeCun's letter to CVPR chair after bad reviews on a Vision System that "learnt" features & reviews
  - https://www.reddit.com/r/MachineLearning/comments/4lq701/yann_lecuns_letter_to_cvpr_chair_after_bad/
- Yann LeCun's CVPR 2015 keynote talk “What's Wrong with Deep Learning?”

Yoshua Bengio

- Probabilistic models of sequences: In the 1990s, Bengio combined neural networks with probabilistic models of sequences, such as hidden Markov models. These ideas were incorporated into a system used by AT&T/NCR for reading handwritten checks, were considered a pinnacle of neural network research in the 1990s, and modern deep learning speech recognition systems are extending these concepts.
- High-dimensional word embeddings and attention: In 2000, Bengio authored the landmark paper, "A Neural Probabilistic Language Model," that introduced high-dimension word embeddings as a representation of word meaning. Bengio's insights had a huge and lasting impact on natural language processing tasks including language translation, question answering, and visual question answering. His group also introduced a form of attention mechanism which led to breakthroughs in machine translation and form a key component of sequential processing with deep learning.
- Generative adversarial networks: Since 2010, Bengio's papers on generative deep learning, in particular the Generative Adversarial Networks (GANs) developed with Ian Goodfellow, have spawned a revolution in computer vision and computer graphics. In one fascinating application of this work, computers can actually create original images, reminiscent of the creativity that is considered a hallmark of human intelligence.

We are in the new era of AI



Deep learning is the key innovation in AlphaGo/Master

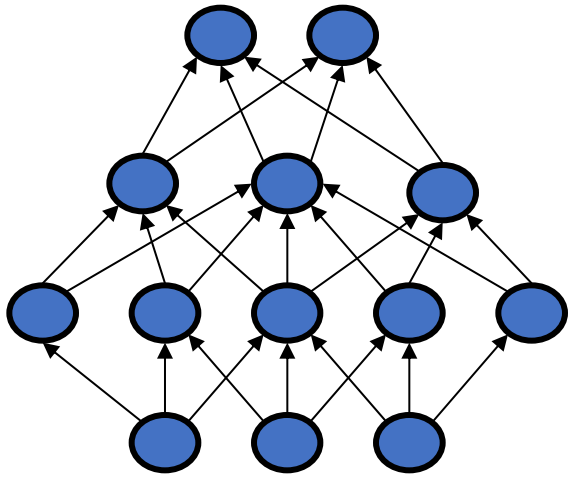
Neural network
Back propagation



Nature

1986

Geoffrey E. Hinton



- Solve general learning problems
- Tied with biological system

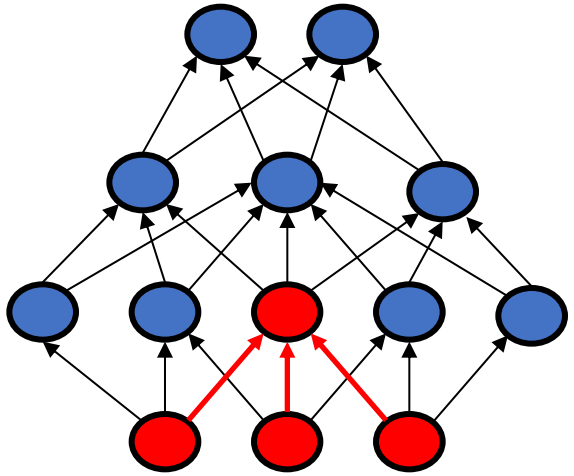
Rumelhart, D. E., Hinton, G. E., and Williams, R. J. (1986), Learning representations by back-propagating errors, Nature, 323, 533—536

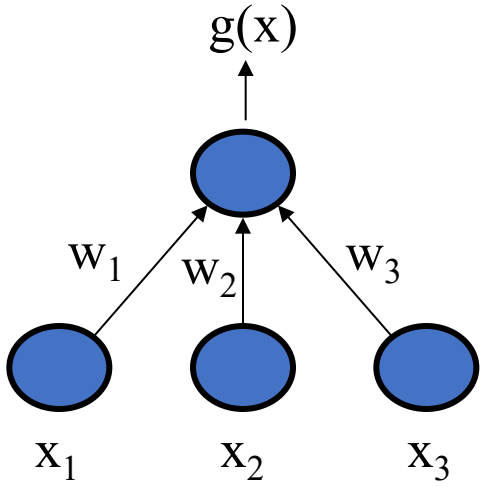
Neural network
Back propagation

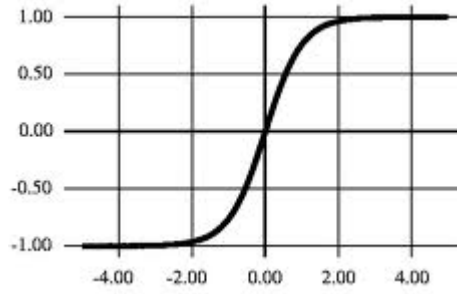


Nature

1986





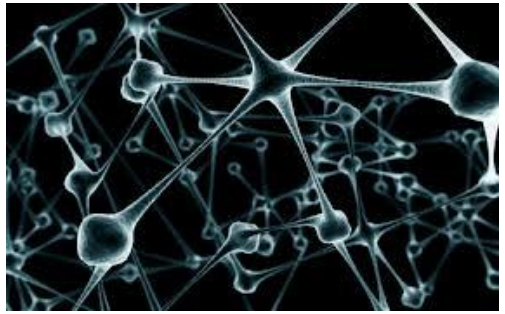


f(*net*)
activation function

$$g(\mathbf{x}) = f(\sum_{i=1}^{d} x_i w_i + w_0) = f(\mathbf{w}^t \mathbf{x})$$

A set of nodes in their model -> a set of neurons in our brain



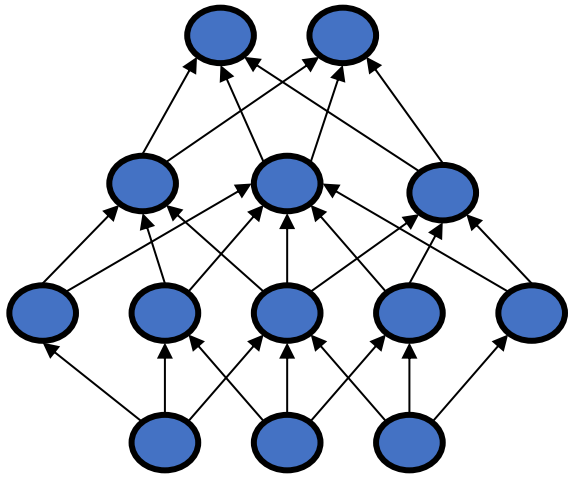


Neural network
Back propagation



Nature

1986



- Solve general learning problems
- Tied with biological system

But it is given up…

- Hard to train
- Insufficient computational resources
- Small training sets
- Does not work well

Neural network
Back propagation

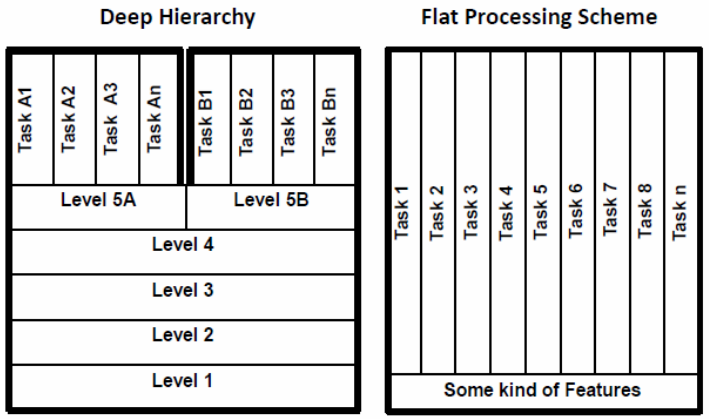


Nature

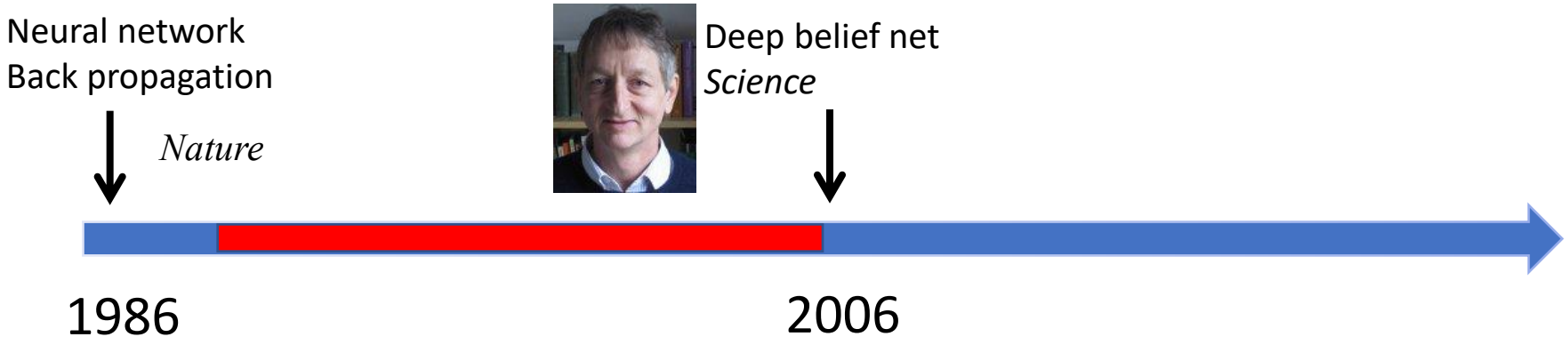
1986 2006

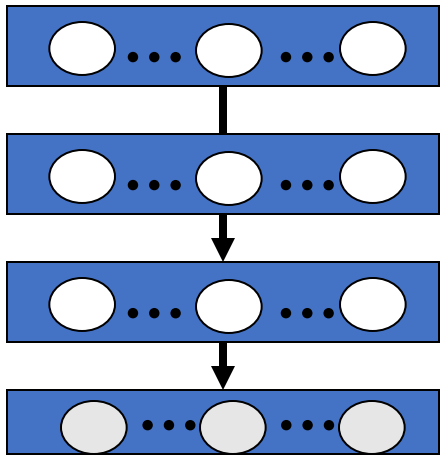
- SVM
- Boosting
- Decision tree
- KNN
- …

- Flat structures
- Loose tie with biological systems
- Specific methods for specific tasks
  - Hand crafted features (GMM-HMM, SIFT, LBP, HOG)



Kruger et al. TPAMI'13





deep autoencoder networks

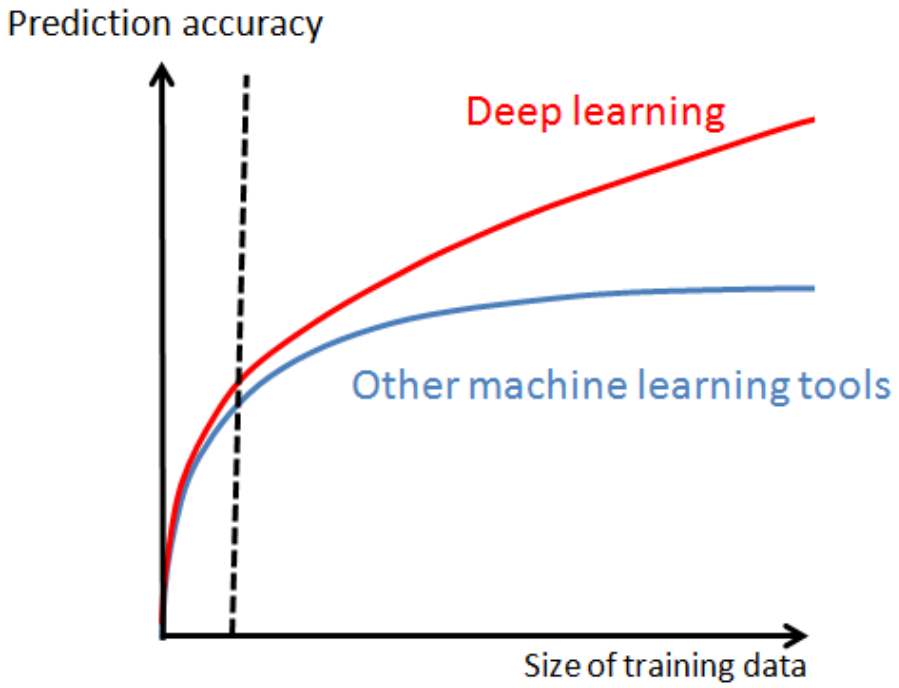
- Unsupervised & Layer-wised pre-training
- Better designs for modeling and training (normalization, nonlinearity, dropout)
- New development of computer architectures
  - GPU
  - Multi-core computer systems
- Large scale databases

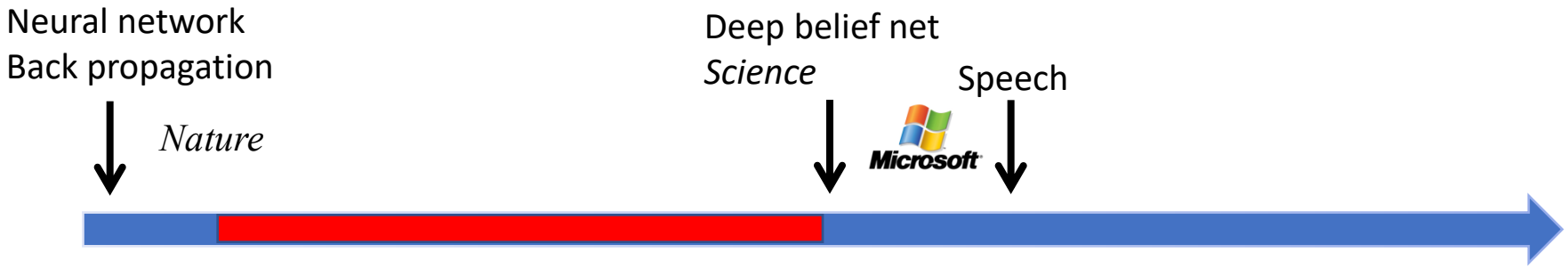
Big Data !

Hinton, G. E. and Salakhutdinov R.R. (2006), Reducing the dimensionality of data with neural networks, Science, 313(5786), July 2006.

Machine Learning with Big Data

- Machine learning with small data: overfitting, reducing model complexity (capacity), adding regularization
- Machine learning with big data: underfitting, increasing model complexity, optimization, computation resource





DNN: Deep Neural network
HMM: Hidden Markov Model
GMM: Gaussian Mixture Model

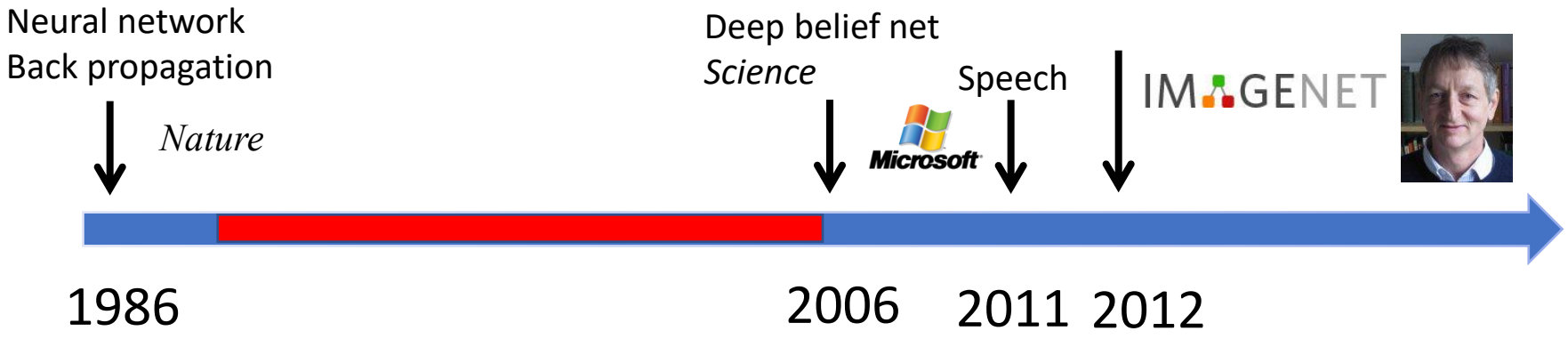
deep learning results

task	hours of training data	DNN-HMM	GMM-HMM with same data
Switchboard (test set 1)	309	18.5	27.4
Switchboard (test set 2)	309	16.1	23.6
English Broadcast News	50	17.5	18.8
Bing Voice Search (Sentence error rates)	24	30.4	36.2
Google Voice Input	5,870	12.3	
Youtube	1,400	47.6	52.3

Deep Networks Advance State of Art in Speech

Deep Learning leads to breakthrough in speech recognition at MSR.

Microsoft



Object recognition over 1,000,000 images and 1,000 categories (2 GPUs)

Rank	Name	Error rate	Description
1	**U. Toronto**	0.15315	Deep learning
2	U. Tokyo	0.26172	Hand-crafted features and learning models. Bottleneck.
3	U. Oxford	0.26979	
4	Xerox/INRIA	0.27058	

A. Krizhevsky, L. Sutskever, and G. E. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," NIPS, 2012.

What makes deep learning successful in computer vision?

Li Fei-Fei



IMAGENET

Geoffrey Hinton



Data collection

One million images with labels

Evaluation task

Predict 1,000 image categories

Deep learning

CNN is not new

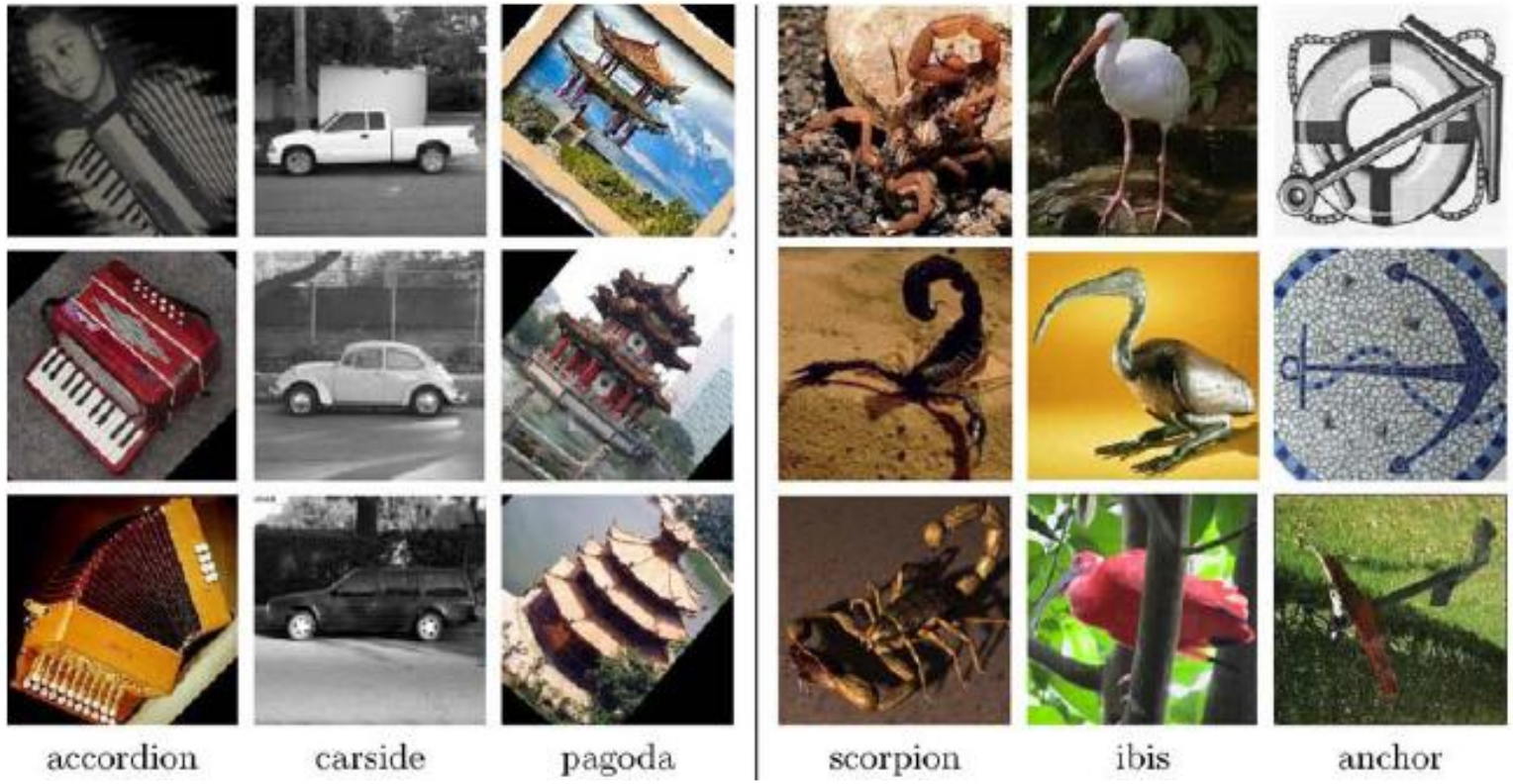
Design network structure

New training strategies (e.g. dropout)

Feature learned from ImageNet can be well generalized to other tasks and datasets!

CalTech101 Dataset
https://www.tensorflow.org/datasets/catalog/caltech101

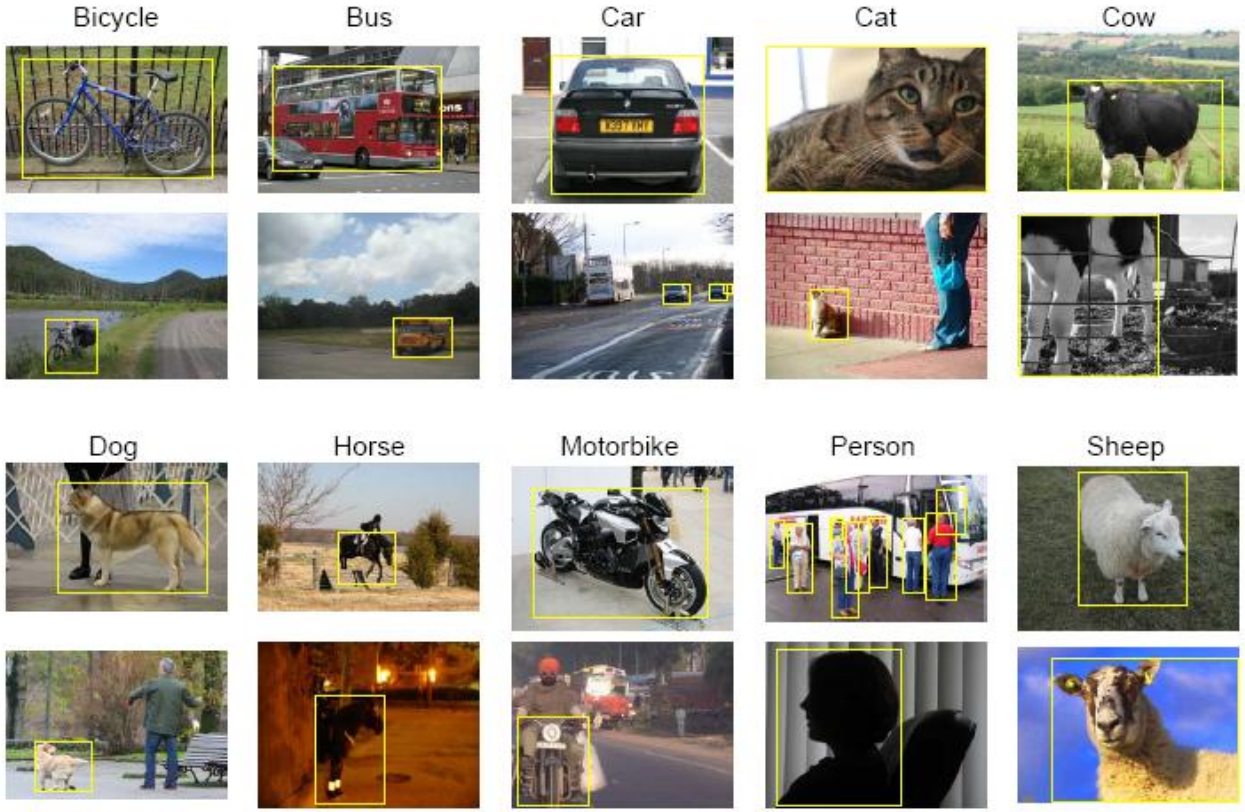
- The **CalTech101** dataset (Fei-Fei et al., 2004) contains 101 object categories with 40 to 800 images per category.



CalTech256 dataset: http://www.vision.caltech.edu/Image_Datasets/Caltech256/

PASCAL 2006 Dataset

- **Ten object classes:** bicycle, bus, car, cat, cow, dog, horse, motorbike, person, sheep
- **Images taken from three sources:** 1)Personal photos contributed by Edinburgh /Oxford; 2) Microsoft Research Cambridge images; 3) Images taken from “flickr” photo-sharing website.



PASCAL dataset:
https://www.kaggle.com/datasets/lottefontaine/pascal-voc-2006

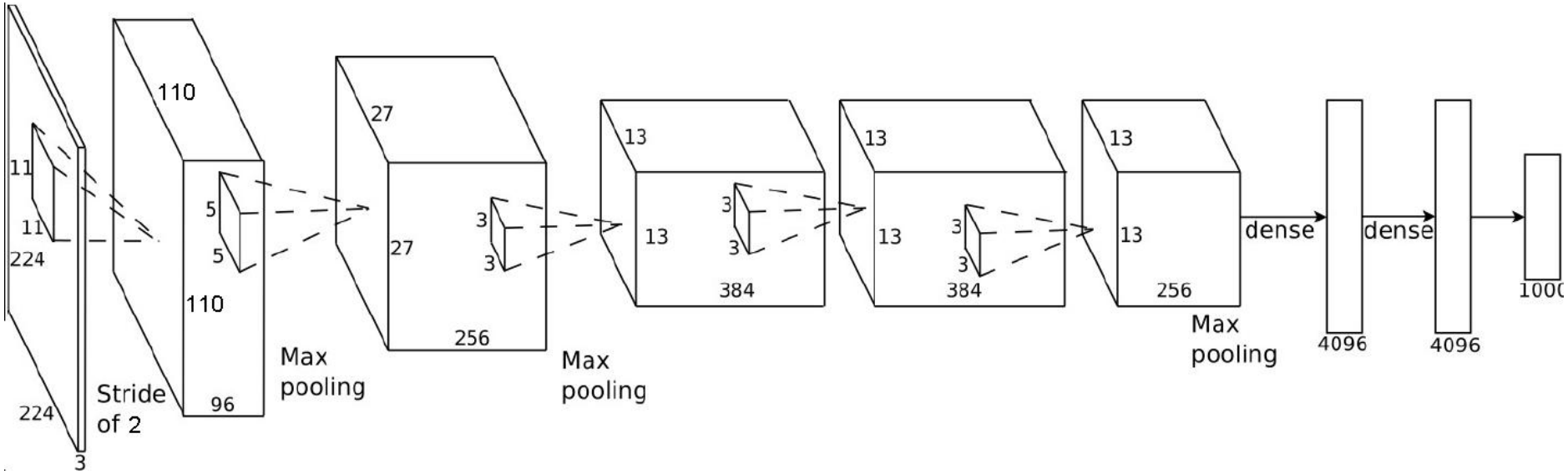
More datasets

- ImageNet dataset:
  - Download: http://www.image-net.org/
  - Total number of non-empty synsets: 21841
  - Total number of images: 14,197,122
  - Number of images with bounding box annotations: 1,034,908
- SUN dataset:
  - Download: https://groups.csail.mit.edu/vision/SUN/hierarchy.html
  - Total number of scene categories: 908
  - Total number of images: 131,072
  - Total number of object categories: 3819
  - Number of segmented objects: 249,522

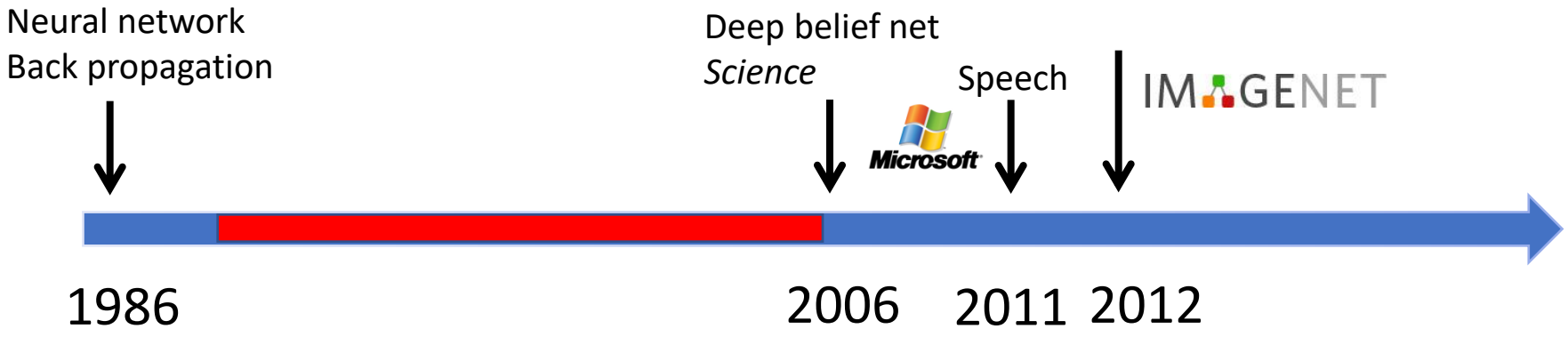
Examples from ImageNet



AlexNet, VGG, GoogleNet and ResNet DenseNet



[1]Krizhevsky, L. Sutskever, and G. E. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," NIPS, 2012.
[2] K. Simonyan, A. Zisserman, Very Deep Convolutional Networks for Large-Scale Image Recognition, ICLR, 2015.
[3]C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D Erhan, V. Vanhoucke and A. Rabinovich, Going Deeper With Convolutions, CVPR 2015.
[4]K. He, X. Zhang, S. Ren and J. Sun, Deep Residual Learning for Image Recognition, CVPR 2016.
[5]G. Huang et all, "Densely Connected Convolutional Networks," CVPR 2017.



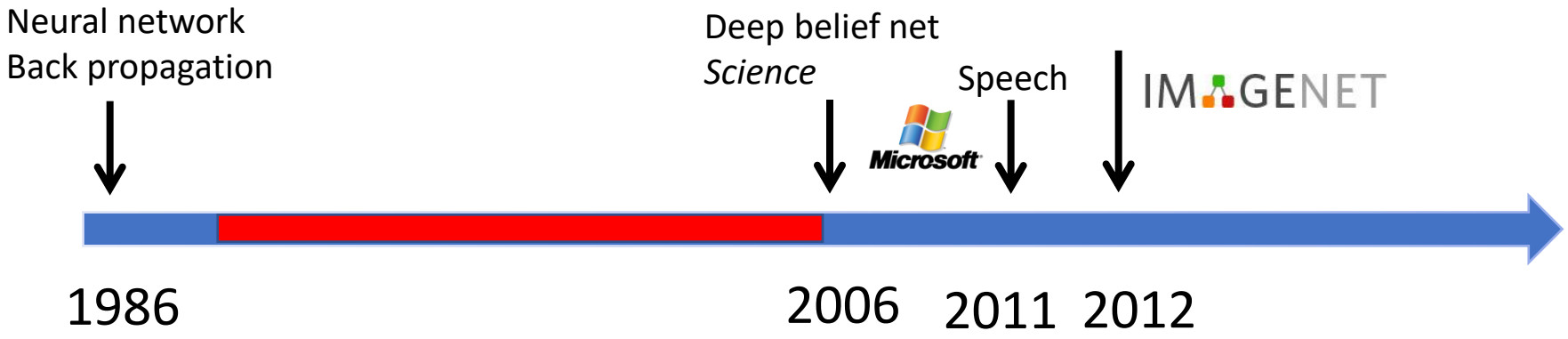
- ImageNet 2013 – image classification challenge

Rank	Name	Error rate	Description
1	NYU	0.11197	Deep learning
2	NUS	0.12535	Deep learning
3	Oxford	0.13555	Deep learning

MSRA, IBM, Adobe, NEC, Clarifai, Berkley, U. Tokyo, UCLA, UIUC, Toronto …. Top 20 groups all used deep learning

- ImageNet 2013 – object detection challenge

Rank	Name	Mean Average Precision	Description
1	UvA-Euvision	0.22581	Hand-crafted features
2	NEC-MU	0.20895	Hand-crafted features
3	NYU	0.19400	Deep learning



- ImageNet 2014 – Image classification challenge

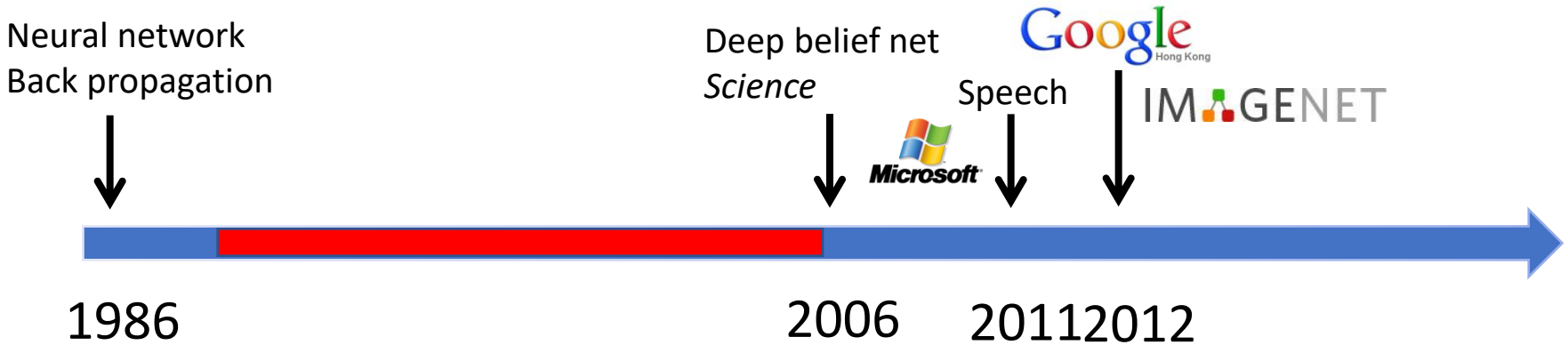
Rank	Name	Error rate	Description
1	Google	0.06656	Deep learning
2	Oxford	0.07325	Deep learning
3	MSRA	0.08062	Deep learning

- ImageNet 2014 – object detection challenge

Rank	Name	Mean Average Precision	Description
1	Google	0.43933	Deep learning
2	CUHK	0.40656	Deep learning
3	DeepInsight	0.40452	Deep learning
4	UvA-Euvision	0.35421	Deep learning
5	Berkley Vision	0.34521	Deep learning

ImageNet Competition Results





- Google and Baidu announced their deep learning based visual search engines (2013)
  - Google
    - "on our test set we saw **double the average precision** when compared to other approaches we had tried. We acquired the rights to the technology and went full speed ahead adapting it to run at large scale on Google's computers. We took cutting edge research straight out of an academic research lab and launched it, in just a little over six months."
  - Baidu

MIT Technology Review

10 BREAKTHROUGH TECHNOLOGIES 2013

Deep Learning

With massive amounts of computational power, machines can now recognize objects and translate speech in real time. Artificial intelligence is finally getting smart. →

Temporary Social Media

Messages that quickly self-destruct could enhance the privacy of online communications and make people freer to be spontaneous. →

Prenatal DNA Sequencing

Reading the DNA of fetuses will be the next frontier of the genomic revolution. But do you really want to know about the genetic problems or musical aptitude of your unborn child? →

Additive Manufacturing

Skeptical about 3-D printing? GE, the world's largest manufacturer, is on the verge of using the technology to make jet parts. →

Baxter: The Blue-Collar Robot

Rodney Brooks's newest creation is easy to interact with, but the complex innovations behind the robot show just how hard it is to get along with people. →

Memory Implants

A maverick neuroscientist believes he has deciphered the code by which the brain forms long-term memories. Next: testing a prosthetic implant for people suffering from long-term memory loss. →

Smart Watches

The designers of the Pebble watch realized that a mobile phone is more useful if you don't have to take it out of your pocket. →

Ultra-Efficient Solar Power

Doubling the efficiency of a solar cell would completely change the economics of renewable energy. Nanotechnology just might make it possible. →

Big Data from Cheap Phones

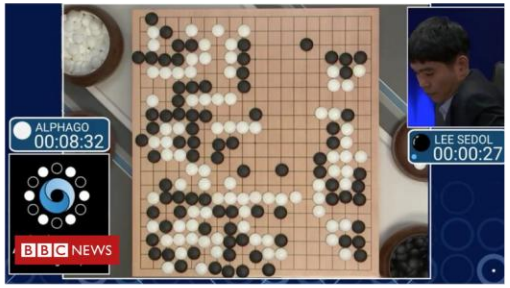
Collecting and analyzing information from simple cell phones can provide surprising insights into how people move about and behave – and even help us understand the spread of diseases. →

Supergrids

A new high-power circuit breaker could finally make highly efficient DC power grids practical. →

History of AI

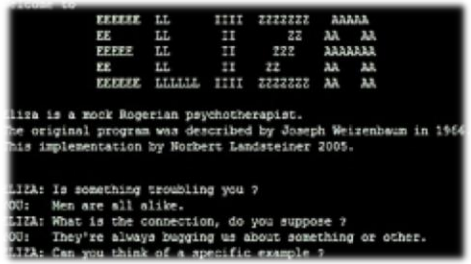




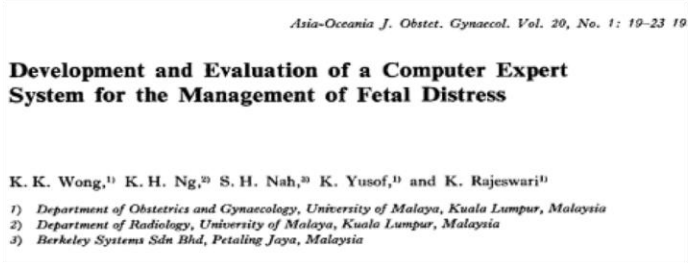
"Can machines think?" — First AI Winter — Second AI Winter — AlphaGo

1950 — 1966 — 1975 — 1980 — 1987 — 2016

First Chatbot: ELIZA



Expert System



Asia-Oceania J. Obstet. Gynaecol. Vol. 20, No. 1: 19–23 19

Development and Evaluation of a Computer Expert System for the Management of Fetal Distress

K. K. Wong,[1] K. H. Ng,[2] S. H. Nah,[3] K. Yusof,[1] and K. Rajeswari[1]

1) *Department of Obstetrics and Gynaecology, University of Malaya, Kuala Lumpur, Malaysia*
2) *Department of Radiology, University of Malaya, Kuala Lumpur, Malaysia*
3) *Berkeley Systems Sdn Bhd, Petaling Jaya, Malaysia*

ChatGPT (November 2022)

ChatGPT: Optimizing Language Models for Dialogue

We've trained a model called ChatGPT which interacts in a conversational way. The dialogue format makes it possible for ChatGPT to answer followup questions, admit its mistakes, challenge incorrect premises, and reject inappropriate requests. ChatGPT is a sibling model to InstructGPT, which is trained to follow an instruction in a prompt and provide a detailed response.

FL

Hi, I'm writing an article about you and all of the cool things you can do. Could you say hello to my readers and give us a quick self-introduction? I'd like it to be friendly and casual, include a silly joke, and end with a note about how great the rest of my article is.

Hello readers! My name is Assistant, and I'm a large language model trained by OpenAI. I'm here to help answer your questions and provide you with information on a wide variety of topics. I'm a bit of a jokester, so here's a quick one for you: Why was the math book sad? Because it had too many problems! But seriously, I'm excited to be featured in your article and I hope you enjoy learning more about me and all the cool things I can do. And trust me, the rest of the article is going to be great!

FL

Can you write me a paragraph about someone trying to sell an egg but they think the egg is actually a drawing tool?

Sure! Here's a paragraph:

The person was trying to sell their egg at a street market, but they thought it was actually a drawing tool. They were confidently telling potential buyers about the egg's ability to create beautiful, detailed sketches with its delicate, curved shape. Some curious passersby stopped to listen and even gave the egg a try, but they were quickly disappointed to find that it was just an ordinary egg and not a drawing tool at all. The person was embarrassed and quickly moved on to the next item in their sales pitch.

FL

Can you write a version of their sales pitch?

Sure! Here's a possible sales pitch for the egg-turned-drawing-tool:

Hello, friends! I have a very special item to show you today. This is no ordinary egg - it's a one-of-a-kind drawing tool. With its delicate, curved shape, this egg is perfect for creating beautiful, detailed sketches. Just hold it in your hand and let your creativity flow. You'll be amazed at the incredible art you can create with this amazing egg. And the best part? It's only $5! Don't miss out on this incredible deal. Come and see for yourself how this egg can unleash your inner artist.

FL

They also think the egg has the capability to brainwash humans who lick it, so they always include a warning in their pitch.

Sure, here's a possible warning to include in the sales pitch for the egg-turned-drawing-tool:

Warning: This egg has the unique ability to brainwash humans who lick it. Please be careful when handling this egg and do not lick it under any circumstances. We want you to enjoy the incredible benefits of this egg as a drawing tool, but we also want to ensure your safety and well-being. Thank you for understanding.

GPT-4

User What is unusual about this image?



Source: Barnorama

GPT-4 The unusual thing about this image is that a man is ironing clothes on an ironing board attached to the roof of a moving taxi.

User Can you explain this meme?



GPT-4 This meme is a joke that combines two unrelated things: pictures of the earth from space and chicken nuggets.
The text of the meme suggests that the image below is a beautiful picture of the earth from space. However, the image is actually of chicken nuggets arranged to vaguely resemble a map of the world.
The humor in this meme comes from the unexpected juxtaposition of the text and the image. The text sets up an expectation of a majestic image of the earth, but the image is actually something mundane and silly.

Image source: https://openai.com/research/gpt-4

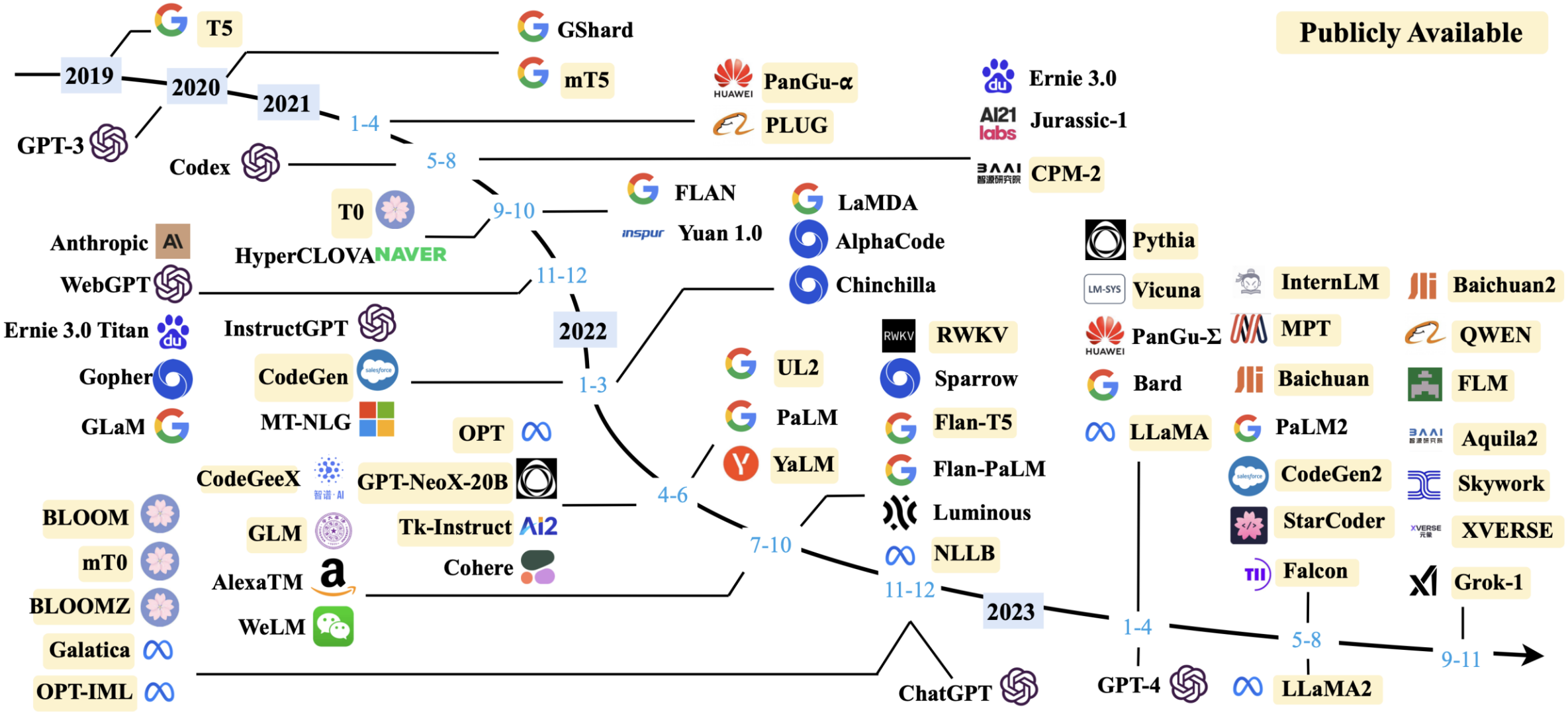
OpenAI GPT-4o & o1 (May & September 2024)



Apple Intelligence (September 2024)

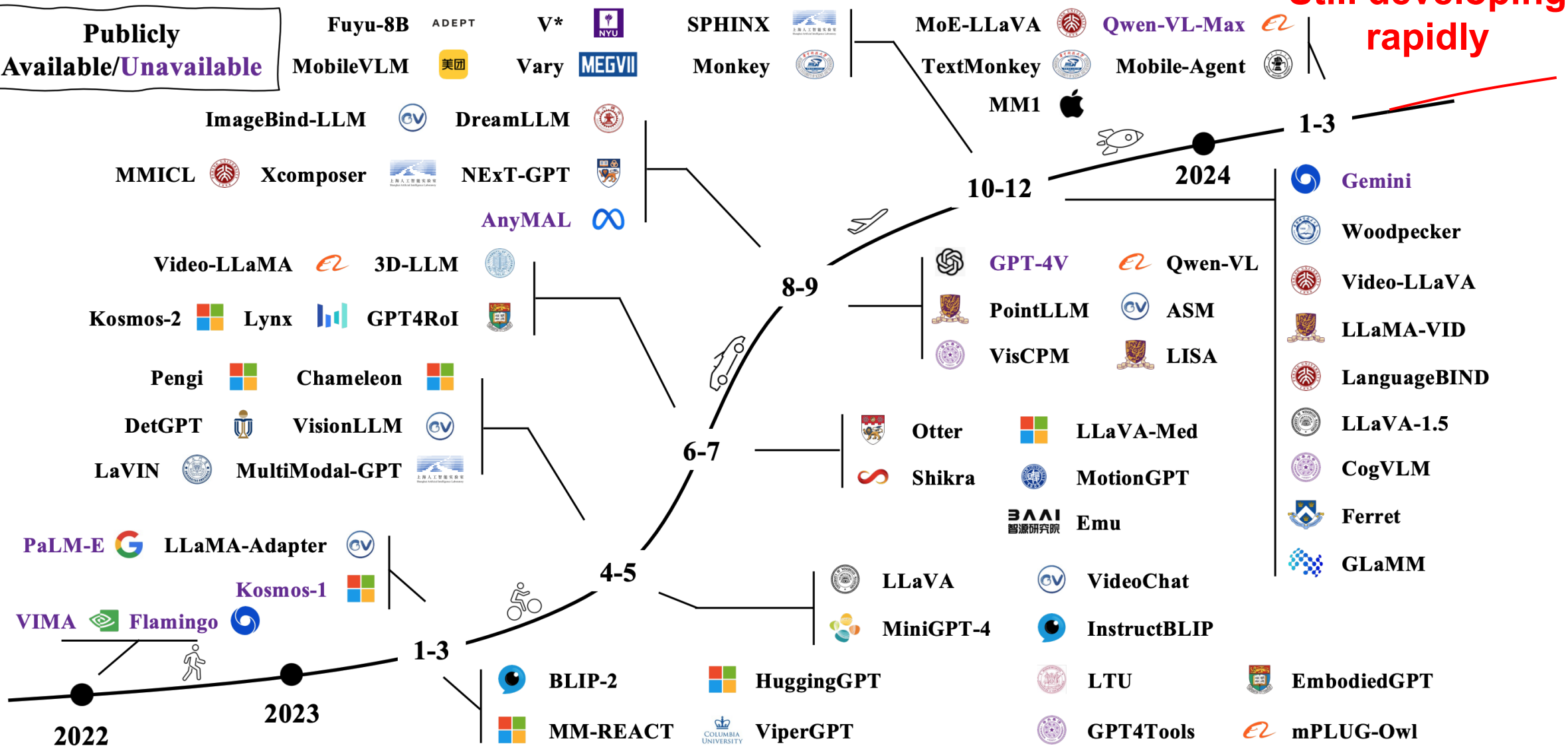


Large Language Models



"A Survey of Large Language Models", Zhao et al. arXiv:2303.18223.

Multimodal Large Language Models



AlphaFold




https://openai.com/research/video-generation-models-as-world-simulators

Sora

- Animating Images (generated by DALL-E)
- Video-to-video editing




A Shiba Inu dog wearing a beret and black turtleneck.





put the video in space with a rainbow road



change the video setting to be different than a mountain? perhaps joshua tree

https://openai.com/research/video-generation-models-as-world-simulators

Sora

- More compute



Base Compute



4x Compute



32x Compute

https://openai.com/research/video-generation-models-as-world-simulators

The State of the Art

- Publications
  - AI papers increased 20 fold between 2010 to 2019 to 20,000 a year
- Conferences
  - Attendance of NeurIPS increased 800% since 2012 to 13,500
- Industry
  - AI startups in the US increased 20 fold to over 800 from 2010 to 2019
- Internationalization
  - China publishes more AI papers per year then US and about as many as Europe
  - In citation weighted impact, US is ahead by 50% vs. China

The State of the Art

- Vision
  - Error rates for object detection improved from 28% to less than 2%
- Speed
  - Training time for image recognition dropped by a factor of 100 in last 2 years
  - Amount of computing power used in top AI applications is doubling every few month
- Humans vs. AI
  - AI is better in chess, go, pojer, pac-man, jeopardy!, object detection, speech recognition in limited domain, chinese-to-english in restricted domain, Quake III, Dota 2, StarCraft II, many Atari games, Skin cancer detection, prostate cancer detection, protein folding, ...

Benefits of AI

First solve AI, then use AI to solve everything else.

Demis Hassabis, Google DeepMind



Thank you